

The University of Azad Jammu and Kashmir, Muzaffarabad.



Computer Networks

Students Name:	Syed Abdulrehman Naseem
Roll No:	2024-SE-18
Course Code:	CS-2203
Lab Number:	09
Instructor:	Engr. Asma Javed

Department of Software Engineering

Lab 9 - Inter-VLAN Routing Using Router-on-a-Stick

1. Lab Overview

This laboratory demonstrates Inter-VLAN Routing using the Router-on-a-Stick method. The lab manual specifies a company with two departments: Development (DEV) and Human Resources (HR). DEV uses VLAN 10 and HR uses VLAN 20; each department has two PCs connected to one switch, and a router provides communication between the VLANs.

1.1 Lab Objectives

- Create the required topology in Cisco Packet Tracer.
- Create VLAN 10 for DEV and VLAN 20 for HR.
- Assign the appropriate PCs to their VLANs.
- Configure the switch-router connection as a trunk.
- Configure router subinterfaces for VLAN 10 and VLAN 20.
- Assign IP addresses and default gateways.
- Test communication between the VLANs using ping.

2. Requirements and Network Design

The required hardware is:

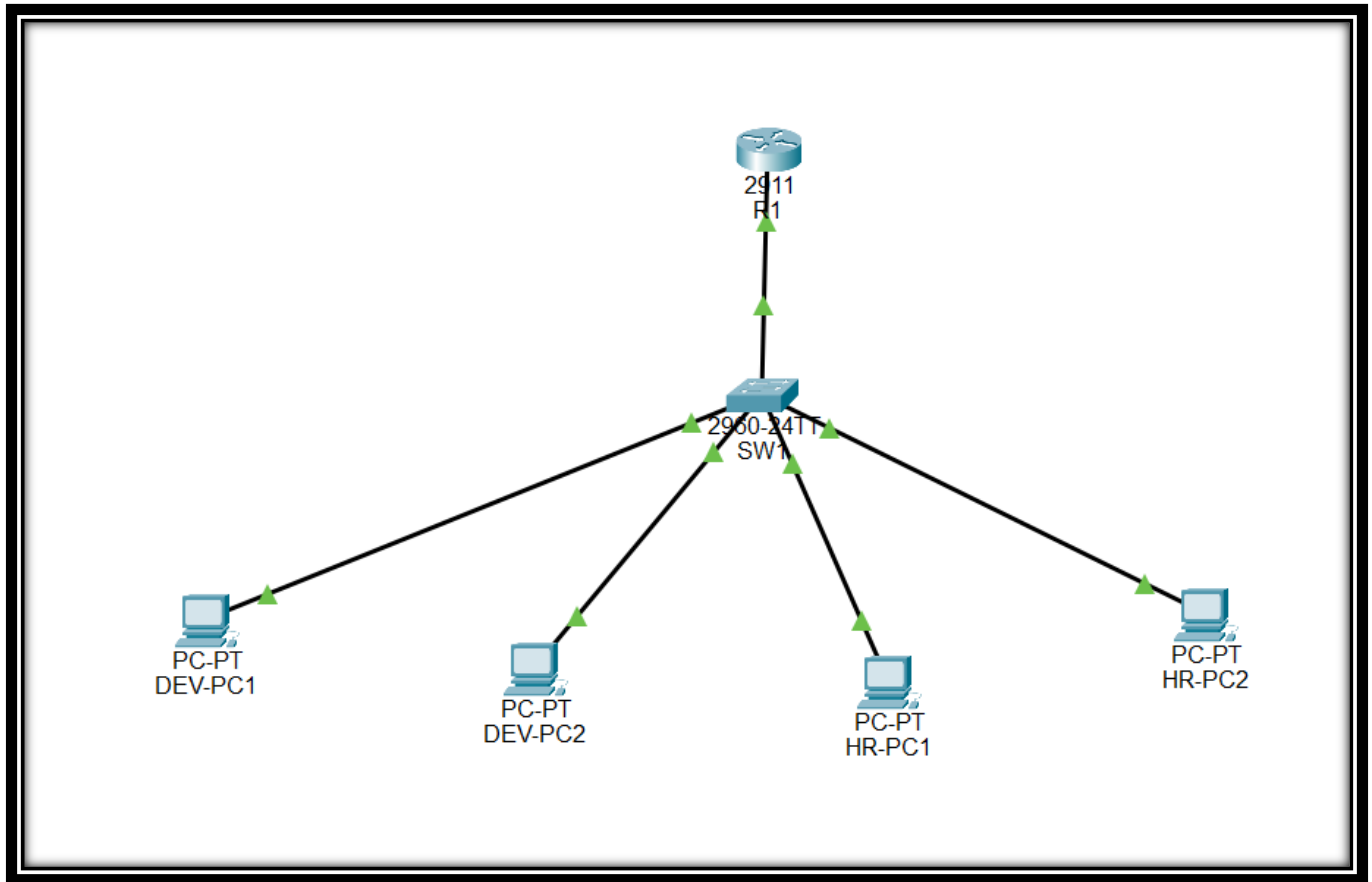
Device	Quantity	Purpose
Router	1	Inter-VLAN routing
Switch	1	VLAN creation and trunk/access connectivity
PC	4	Two DEV hosts and two HR hosts

VLAN and IP Addressing Plan

Department	VLAN	Network	Default Gateway
Development (DEV)	10	192.168.10.0/24	192.168.10.1
Human Resources (HR)	20	192.168.20.0/24	192.168.20.1

TASK 1 — Topology Creation

Creating the topology in Cisco Packet Tracer. Place 1 router, 1 switch, and 4 PCs. Connect two PCs to the switch for DEV and two PCs for HR. Connect the switch to the router using an Ethernet link.



Suggested device labels: R1, SW1, DEV-PC1, DEV-PC2, HR-PC1, HR-PC2.

3. Physical and Logical Connections

Using copper straight-through Ethernet connections for PC-to-switch links and the switch-to-router link. The switch-router link will later operate as an 802.1Q trunk so that traffic for both VLAN 10 and VLAN 20 can cross the single physical link.

TASK 2 — Cabling

Connect DEV-PC1 and DEV-PC2 to access ports assigned to VLAN 10, HR-PC1 and HR-PC2 to access ports assigned to VLAN 20, and connect SW1 to R1. Record the actual interfaces used.

4. VLAN Creation and Port Assignment

VLAN 10 is named DEV and VLAN 20 is named HR. The two DEV PCs are placed in VLAN 10, while the two HR PCs are placed in VLAN 20. The port connected to the router is configured as a trunk.

Example Switch Configuration

```
enable
```

```
configure terminal
```

```
hostname SW1
```

```
vlan 10
```

```
name DEV
```

```
exit
```

```
vlan 20
```

```
name HR
```

```
exit
```

```
interface range fastEthernet 0/1-2
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
exit
```

```
interface range fastEthernet 0/3-4
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

```
interface fastEthernet 0/24
```

```
switchport mode trunk
```

```
no shutdown
```

```
exit
```

end

copy running-config startup-config

TASK 3 — VLAN and Switch Configuration

show vlan brief

show interfaces trunk

```
SW1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	DEV	active	Fa0/1, Fa0/2
20	HR	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1#
```

Copy Paste

Top

```

SW1#show interfaces fastEthernet 0/24 status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/24    connected trunk      a-full  a-100  10/100BaseTX

SW1#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/24    1-1005

Port      Vlans allowed and active in management domain
Fa0/24    1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    1,10,20

SW1#

```

Copy

Paste

5. Router-on-a-Stick Configuration

Router-on-a-Stick uses one physical router interface divided into logical subinterfaces. Each subinterface is associated with one VLAN using 802.1Q encapsulation and acts as the default gateway for that VLAN.

Example Router Configuration

enable

configure terminal

hostname R1

interface gigabitEthernet 0/0

no shutdown

exit

interface gigabitEthernet 0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

exit

interface gigabitEthernet 0/0.20

```

encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit

end
copy running-config startup-config

```

TASK 4 — Router Configuration

Configuring the router using the interface connected to the switch. If your router uses a different interface name, use the interface actually connected to SW1. Verify the subinterfaces with show ip interface brief.

```
show ip interface brief
```

```

R1#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 unassigned      YES unset   up          up
GigabitEthernet0/0.10 192.168.10.1    YES manual  up          up
GigabitEthernet0/0.20 192.168.20.1    YES manual  up          up
GigabitEthernet0/1    unassigned      YES unset   administratively down down
GigabitEthernet0/2    unassigned      YES unset   administratively down down
Vlan1               unassigned      YES unset   administratively down down
R1#

```

Copy Paste

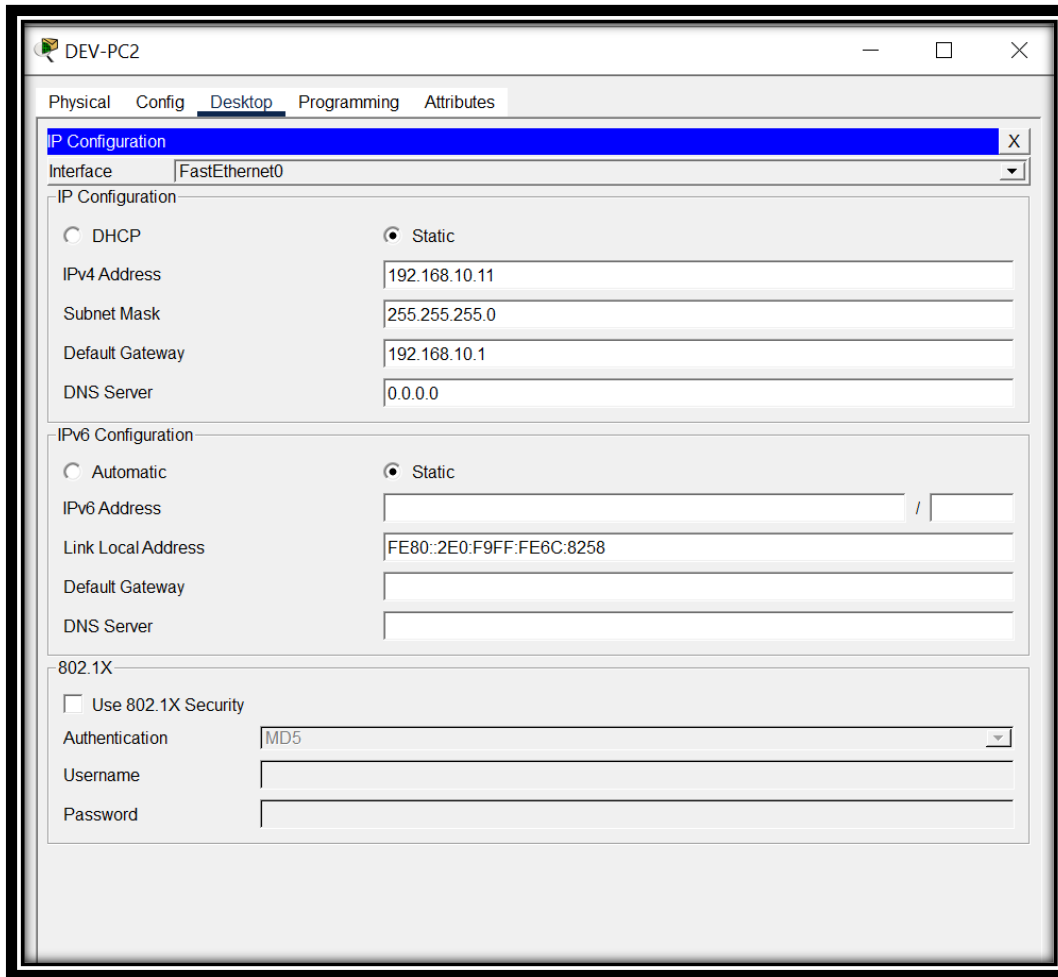
Top

6. End-Device IP Addressing

Device	VLAN	IP Address	Subnet Mask	Default Gateway
DEV-PC1	10	192.168.10.10	255.255.255.0	192.168.10.1
DEV-PC2	10	192.168.10.11	255.255.255.0	192.168.10.1
HR-PC1	20	192.168.20.10	255.255.255.0	192.168.20.1
HR-PC2	20	192.168.20.11	255.255.255.0	192.168.20.1

TASK 5 — Configuring PC IP Addresses

For each PC, IP Configuration is performed and the values from the table above.



The screenshot shows the 'DEV-PC2' window with the 'Desktop' tab selected. The 'IP Configuration' window is open, showing the 'FastEthernet0' interface. The 'IP Configuration' section has 'Static' selected, with the following values: IPv4 Address: 192.168.10.11, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.10.1, and DNS Server: 0.0.0.0. The 'IPv6 Configuration' section has 'Static' selected, with the following values: IPv6 Address: (empty), Link Local Address: FE80::2E0:F9FF:FE6C:8258, Default Gateway: (empty), and DNS Server: (empty). The '802.1X' section has 'Use 802.1X Security' unchecked, with 'Authentication' set to 'MD5', and 'Username' and 'Password' fields are empty.

7. Connectivity Testing

The expected result is successful communication between VLAN 10 and VLAN 20 through the router. Ping tests should be performed after both VLANs, the trunk, router subinterfaces, and PC addressing have been configured.

TASK 6 — Ping Tests

From DEV-PC1, test the local gateway and an HR host:

```
ping 192.168.10.1
```

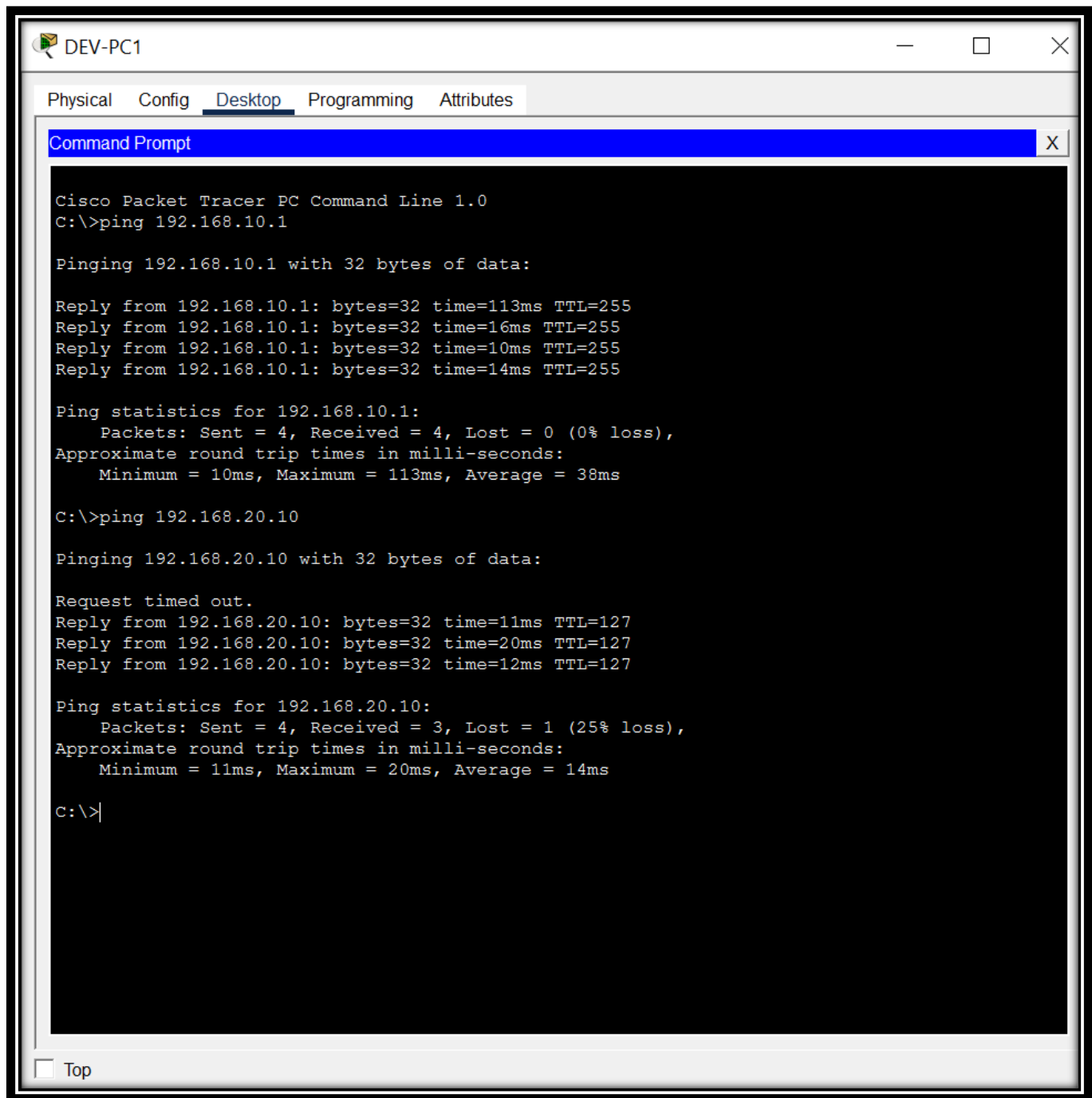
```
ping 192.168.20.10
```

From HR-PC1, test the local gateway and a DEV host:

ping 192.168.20.1
ping 192.168.10.10

Record the actual results below. A first ping may occasionally time out while ARP information is learned; confirm that subsequent packets succeed before declaring connectivity successful.

Source	Destination	Command	Actual Result
DEV-PC1	192.168.10.1	ping	SUCCESS
DEV-PC1	192.168.20.10	ping	SUCCESS
HR-PC1	192.168.20.1	ping	SUCCESS
HR-PC1	192.168.10.10	ping	SUCCESS



8. Verification Commands

SW1:

show vlan brief

show interfaces trunk

R1:

show ip interface brief

PCs:

ping <destination-IP>

TASK 7 — Verification Evidence

```
SW1>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Gig0/1 Gig0/2
10	DEV	active	Fa0/1, Fa0/2
20	HR	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1>show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/24	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/24	1-1005

Port	Vlans allowed and active in management domain
Fa0/24	1,10,20

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/24	1,10,20

```
SW1>
```

Copy

Paste

Top

9. Results and Discussion

The completed design separates the Development and Human Resources departments into VLAN 10 and VLAN 20. The switch provides Layer-2 VLAN separation, while the router's 802.1Q subinterfaces provide Layer-3 gateways for the two networks. The trunk carries traffic from both VLANs between the switch and router. Successful ping tests between a DEV host and an HR host demonstrate inter-VLAN communication through the router.

10. Conclusion

This laboratory demonstrated Inter-VLAN Routing using the Router-on-a-Stick approach. VLAN 10 was used for Development and VLAN 20 for Human Resources. The switch was configured with access ports for the respective departments and a trunk toward the router. Router subinterfaces provided the default gateways for both networks. After assigning appropriate IP addresses and gateways to the PCs, connectivity between the two VLANs was verified using ping tests.